SMARTLEARN AI - ACADEMIC SYSTEM

Program: BEng Mechanical Engineering | Course Materials

Course Overview & Prerequisites

Thermodynamics is the science of heat, work, and the transformations of energy. This course covers the fundamentals, including properties of pure substances, the First and Second Laws of Thermodynamics, entropy, exergy analysis, and gas/vapor power cycles. Prerequisites: Engineering Physics, Calculus II.

First Law of Thermodynamics

The First Law of Thermodynamics is a statement of the conservation of energy principle. It asserts that energy can neither be created nor destroyed; it can only change forms.

For a closed system undergoing a cycle, the net heat added to the system is equal to the net work done by the system:

Q - W = Delta U

Where:

Q = Net heat transfer

W = Net work done

Delta U = Change in internal energy of the system.

Second Law of Thermodynamics & Entropy

The First Law places no restriction on the direction of energy flow. The Second Law establishes that processes occur in a certain direction and that energy has quality as well as quantity.

Key statements:

- Clausius Statement: It is impossible to construct a device that operates in a cycle and produces no other effect than the transfer of heat from a lower-temperature body to a higher-temperature body.
- Kelvin-Planck Statement: It is impossible for any device that operates on a thermodynamic cycle to receive heat from a single reservoir and produce a net amount of work.